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# Heuristic and random search algorithm in optimization of route planning for Robot's geomagnetic navigation

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ABSTRACT

To improve the efficiency and accuracy, a new combination algorithm for route planning is proposed, by considering underwater geomagnetic matching navigation area and distribution of environmental constraints. Firstly, with geomagnetic navigation matching regions, Dijkstra algorithm can obtain the primary route points. Secondly, the environmental constraints models are built and normalized, and the route planning environment constrained cost model is established. Thirdly, with the relationships between time, function relation, constraint condition and variable in the environment constrained cost model, the particle swarm optimization algorithm is introduced. With the primary route pints, the route planning is transformed into route optimization. Finally, the primary route points are used as the initial input of the particle swarm optimization algorithm, then the methods of selecting the inertia weight of the particle swarm and the particle coding are improved. The optimal route planning of Dijkstra algorithm and particle swarm optimization is realized. Simulation results demonstrate that the particle size of the search space can get a minimized evaluation, more narrowed search range and higher efficient search. The combination algorithm guarantees the global optimal while ensures the local optimal, then, the non-matching navigation areas can be effectively avoided, and efficient route planning functions can be achieved.

## 1. Introduction

Underwater geomagnetic navigation is a kind of navigation method based on the geomagnetic map with rich underwater geomagnetic characteristic information [1]. By combining the underwater vehicle's position information of Inertial Navigation System (INS) and magnetometer, the geomagnetic characteristic matching algorithm can obtain accurate position information to perform navigation tasks [2,3]. Planning optimal route based on current position and navigation matching region distribution is an urgent issue to be settled in underwater vehicle geomagnetic navigation system. The final aim of planning optimal route is to obtain the shortest route through the adaptation area effectively. The route planning problem is generally considered as a NP-hardness (nondeterministic polynomial time) solving equation, which is a complex function with uncertain polynomials. NP-hard means "at least as hard as any NP-problem", although it might, in fact, be harder. In NP-hardness solving equation, optimum solution cannot be obtained by using the current polynomial time methods. Different optimization algorithms are usually used to obtain the possible optimal solution. Comparing with heuristic algorithm and stochastic algorithm, the optimal algorithm takes the route cost models as fitness function, and achieves better optimization performance according to the evolution of particles. However, the optimal algorithm usually has some deficiencies, such as slow convergence, falling into local optimum, and so on [4]. Matching error of the iterative closest contour point algorithm for terrain-aided navigation, and an investigation into the matching errors of the revised iterative closest contour point (ICCP) algorithm for underwater TAN were presented [5]. Impact of terrain factors on the matching performance of terrain aided navigation, and a new method were proposed to describe the terrain elevations quantitatively by factor analysis for terrain-aided navigation [6].

In underwater geomagnetic navigation, the determination of the optimal track is closely related to the distribution of the geomagnetic adaptation area. On the other hand, the track of the underwater vehicle also needs to be optimized. Track planning is a comprehensive interdisciplinary research topic, and it is also a major component of the aircraft's mission. Track planning is a complex problem that solves nondeterministic polynomials, and the known polynomial time problems cannot be used to find the optimal solution. Optimization algorithms are often used to obtain problem solutions. According to the planning scope, they can be divided into global planning algorithms and local

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optimization algorithms. According to the calculation methods of planning decisions, they can be divided into optimal, heuristic, and random algorithms.

The A * algorithm is an omnidirectional extended search. The characteristics of the objective function are heuristic rules to change the expansion order of nodes. The A * algorithm has a large search space and there are redundant expansion nodes, which requires a lot of storage space and search time. The sparse A * algorithm is an improvement of the A * algorithm. It reduces the number of invalid search points, and controls the extended area from a full range of 8-node search to the fan-shaped area. The sparse A * algorithm is used for trajectory planning to combine heuristic information such as constraints.

The Voronoi diagram algorithm is a spatial segmentation algorithm. It consists of a set of continuous polygons consisting of vertical bisectors that connect straight lines between two neighboring points. It is a geometric structure that represents the approximate information of a set of points or entities. The vector method and grid method are two kinds of Voronoi diagram algorithms.

The geomagnetic navigation track planning of an underwater vehicle needs to comprehensively consider the spatial modeling, constraints, planning algorithms, navigation accuracy, and other factors of the matching area to select the optimal track. The constraint is the key issue in determining the optimal solution space. The core of the trajectory planning problem is to optimize the objective function. The algorithms have their own advantages and disadvantages, and different geomagnetic map features have different degrees of influence on different planning algorithms. A hybrid track planning algorithm that combines multiple algorithm technologies and is suitable for geomagnetic navigation features is one of the research directions of underwater vehicle track planning.

To overcome the deficiencies of route planning optimal algorithm for geomagnetic field navigation, a route combination planning method based on Dijkstra algorithm (Dijkstra's algorithm or Dijkstra's Shortest Path First algorithm is an algorithm for finding the shortest paths between nodes in a graph, which may represent, for example, road networks. It was conceived by computer scientist Edsger W. Dijkstra) and modified Particle Swarm Optimization (PSO) algorithm is proposed. Off-line route planning with geomagnetic feature constraints is carried out by using Dijkstra algorithm. Then the primary route node is treated as the initial input of the PSO to improve the encoding method. The innovation of this method lies in regarding time information as 'particles' in space, and improving the convergence performance. A series of experiments was made to compare the performance between combination planning method and the standard PSO, and demonstrated that the combination planning method has better convergence and powerful ability of global optimization.

## 2. Analysis and modeling of underwater environmental constraints

The traditional route planning [7] is a disposable off-line planning. However, even the vehicle works in matching areas, it needs to consider several factors, involving various obstacles, steering angle and time constraint, etc. The purpose of route planning is to design a shortest route with the algorithm, which can meet the boundary conditions of navigation, maneuver performance, threat avoidance and so on. Underwater vehicle route planning has been proved to be an uncertain polynomial problem with 3D kinematics and dynamics constraint. As the complex underwater environment, solving the problem of route planning is easy to cause the search algorithm to converge too slowly or fall into local convergence. In addition, it is difficult to use terrain aided navigation technology in the flat sea area. Under the constraints of obstacle and prior knowledge, the 2D plane route planning method is suitable for the practical application background of underwater geomagnetic navigation. The key features involved in the analysis and modeling of underwater environment constraints are as follows, geomagnetic intensity, falling gradient, steering angle, obstacle and the length of voyage. As uncertainty of function parameters, there have not definite relationships among them. To simplify the calculation, the simplified route function method is usually used. Taking the different function weights into account, the model of route evaluation function is obtained as

$$\begin{aligned} \min fit\left(x_i, y_i\right) &= \omega_1 J_{terrain}\left(x_i, y_i\right) + \omega_2 J_{slope}\left(x_i, y_i\right) + \omega_3 J_{angle}\left(x_i, y_i\right) \\ &\quad + \omega_4 J_{mon}\left(x_i, y_i\right) + \omega_5 J_{length} \end{aligned} \tag{1}$$

$$s.t. \quad 0 < \omega_1 < 1$$
$$0 < \omega_2 < 1$$
$$0 < \omega_3 < 1$$
$$0 < \omega_4 < 1$$
$$0 < \omega_5 < 1$$
$$\omega_1 + \omega_2 + \omega_3 + \omega_4 + \omega_5 = 1$$

where $fit(\mathrm{x_i}, \mathrm{y_i})$ is the evaluation function of route planning, $\omega_1$, $\omega_2$, $\omega_3$, $\omega_4$ and $\omega_5$ are the weighting factors of each restriction. $\mathrm{J_{terrain}}$ is normalized geomagnetic intensity, $\mathrm{J_{slope}}$ is normalized falling gradient, $\mathrm{J_{angle}}$ is normalized steering angle, $\mathrm{J_{mon}}$ is normalized obstacle and $\mathrm{J_{length}}$ is normalized the length of voyage.

### 2.1. Geomagnetic constraint

The working depth of underwater vehicle is limited. To meet the requirement of safety, the superimposed geomagnetic intensity function in navigation area is given by

$$\mathrm{J_{terrain}} = \begin{cases} \mathrm{f_{safe}} & \text{if } \mathrm{f_{mission}} < \mathrm{f_{safe}} \\ \mathrm{f_{mission}} & \text{else} \end{cases} \tag{2}$$

where $\mathrm{J_{terrain}}$ is the superimposed geomagnetic intensity function for safe navigation, $\mathrm{f_{safe}}$ is the minimum safe sailing depth to protect the underwater vehicle, $\mathrm{f_{mission}}$ is the depth between the vehicle and seabed.

As the magnetic gradient increases, it is more difficult for underwater vehicle to follow the planning route. It is limited when the slope is greater than the maximum angle. S function is adopt to model the following probability $\mathrm{P(i,j)}$.

$$\mathrm{P(i,j)} = \frac{1}{1 + \exp\left[-\mathrm{k}\frac{-\mathrm{f(i,j)}+\mu}{\varphi_s}\right]} \tag{3}$$

$$\mathrm{J_{slope}} = 1 - \mathrm{P(i,j)} \tag{4}$$

where $k$ and $\varphi_s$ are parameters of S function, $f(i,j)$ is the geomagnetic intensity function at node $(i,j)$, $\mu$ is a threshold, $J_{slope}$ is the following probability function of vehicle. Considering the inclination range of vehicle, geomagnetic gradient is usually chosen as the quantitative threat factor.

### 2.2. Maximum steering angle constraint

To avoid the risk of collision caused by sharp turn, the maximum turning angle limits the trajectory to be within or below the maximum turning angle. Supposing the maximum turning angle of the vehicle is $\theta_{\max}$, the vehicle must satisfy the horizontal angle constraint. The maximum steering angle constraint is given by

$$\mathrm{J_{angle}} = \begin{cases} 0 & \theta \le \theta_{\max} \\ \theta - \theta_{\max} & \theta > \theta_{\max} \end{cases} \tag{5}$$

where $\theta$ is the current steering angle of the vehicle which can be calculated by three adjacent route nodes $(\mathrm{x_{i-1}}, \mathrm{y_{i-1}})$, $(\mathrm{x_i}, \mathrm{y_i})$ and $(\mathrm{x_{i+1}}, \mathrm{y_{i+1}})$, $\mathrm{d(i-1,i)}$ is the distance between $\left(\mathrm{x_{i-1}}, \mathrm{y_{i-1}}\right)$ and $\left(\mathrm{x_i}, \mathrm{y_i}\right)$. Then they are calculated by

$$\mathrm{d(i-1,i)} = \sqrt{(\mathrm{x_{i-1}} - \mathrm{x_i})^2 + (\mathrm{y_{i-1}} - \mathrm{y_i})^2} \tag{6}$$

$$\theta = \cos^{-1}\frac{\mathrm{d(i-1,i+1)}^2 - \mathrm{d(i-1,i)}^2 - \mathrm{d(i,i+1)}^2}{2\mathrm{d(i-1,i)\,d(i,i+1)}} \tag{7}$$

### 2.3. Obstacle constraint

It is necessary for vehicle to avoid the static obstacles in the process of moving such as the seabed, coke, rock and dark pile. The risk is greater when the obstacle is closer. So the vehicle must react when sonar detects the relative distance and orientation of obstacles. The obstacles include dangerous terrain and underwater listening here. In the absence of sound propagation speed, sea temperature and other factors, the threat of vehicle is only related to distance. The constraint of obstacles can be equivalent to the underwater acoustic monitoring source. Then the function of each node of underwater acoustic monitoring source can be modeled by Gauss distribution function.

$$J_{mon} = \sum_{j=0}^{n-1} \left\{ \frac{Q_j}{2\pi\sqrt{\det(H_j)}} \cdot \exp[-\frac{1}{2}(r-\mu_j)^T H_j^{-1}(r-\mu_j)] \right\} \tag{8}$$

where $r = [x, y]^T$ is the position vector, Q is the threat level of the source, H is the variance, $\mu_j$ is the position vector of j.

### 2.4. Sailing length constraint

The length of voyage represents the sum of the length from starting node to target node. At the route node $(x_i, y_i)$, $i = 0, 1, 2, \ldots N$, where N is the number of route nodes. Its navigation length function $J_{length}$ can be expressed as

$$J_{length} = \sum_{j=0}^{N-1} \sqrt{(x_{j-1} - x_j)^2 + (y_{j-1} - y_j)^2} \tag{9}$$

## 3. Dijkstra algorithm

Dijkstra algorithm is a classical algorithm to solve the shortest route problem with single source in graph theory [8]. It is the most representative shortest route algorithm. Dijkstra algorithm proposes a method of iteration according to the order of route length. It is widely used in GIS search, data structure, graph theory, operations research and other fields. The basic idea of the Dijkstra algorithm is to generate a root with the fixed starting node as the tree structure, then the route from the root to each node is the shortest route. There is no negative weight between network nodes in traditional Dijkstra algorithm. Whether join a new node to the spanning tree depends on the distance and adjacent relationship. There is a graph G with N different nodes, thus the number of shortest route from any node to another is (N-1). Supposing the starting node is $V_0$, $V = \{V_0 \quad V_1 \quad \cdots \quad V_{n-1}\}$ is used to represent a collection of all vertices in graph G. The ordered element $(V_i, V_j)$ means the arc from $V_i$ to $V_j$. The set of all edges is represented by E, and the weight of each edge is defined by the weight function matrix $E \rightarrow [0, \infty)$, where $|E| = n$. Thus $matrix_{ij}$ is a non negative cost from $V_i$ to $V_j$, represented by weight. The weight can be determined by the distance between two nodes, which is the sum of weights on all sides of the route. The shortest route is calculated by Dijkstra algorithm according to ascending order.

The main steps of the Dijkstra algorithm are described as follows. The nodes are divided into two parts. One is the temporary node set named S, and the other is the unlabeled node set named U. At the beginning, there is only a starting node in set S. Then the node is added to it every time a shortest route is found.

Step 1: Initialize

NodeList is initialized, as well as all node weights list *matrix*. The queue *Heap* is initialized by *Insert* operation, and the source node is inserted. At the same time, Index is set up in which *Heap* corresponds to NodeList.

Step 2: Extract $Node[j]$ which has shortest distance in the priority queue

Take the head node $V_j$ in the queue. Thus $Node[j]$ is the shortest route from the starting node $V_1$ to the end $V_j$. If the node $V_j$ is target, the remaining nodes in the queue *Heap* and their index relations are released. The algorithm is finished.

Step 3: Join the node $V_k$ starting from $V_j$ into the queue

The principle is $dist[k] = min\{matrix_{1k}, dist[j] + matrix_{jk}\}$. Meanwhile, the table Index is updated. If the node $V_k$ has been out of the queue, it is not processed.

Step 4: Repeat step 2 until finding the destination $V_D$.

## 4. Particle swarm optimization algorithm

The basic idea of standard PSO [9,10] is as follows. The bird's predation is usually regarded as a searching problem in space, while each potential solution is a bird. This method takes the bird as an abstract particle [10] without volume or weight. Every particle has an adaptation value determined by the optimization function. The velocity and direction are determined by the velocity vector. It is the information transferred and interacted among particles that leads them move to the target. So the target in the space is the 'food' that the birds are searching for.

PSO can dynamically route the current search condition and adjust the corresponding according to its unique memory [11]. During iteration, each individual can remember its best position currently. Besides, it updates itself by routing two extremums. One is the best position the individual has experienced, known as the 'local optimum', denoted as $P_{id}$. If there is a group of $m$ in the $D$ dimension, $P_{id}$ is the best fitness function value obtained by particle $i$($i = 1, 2, \ldots m$). The other is the optimal location found by all particles in the group, known as the 'global optimum' while $d = 1, 2, \ldots, D$, denoted as $G_{id}$.

The position of particle i in the search space is denoted as $x_i = (x_{i1}, x_{i2}, \ldots x_{iD})$, and the corresponding change of particle position is $v_i = (v_{i1}, v_{i2}, \ldots v_{iD})$. If iterations have been taken $t$ times, the position and velocity of each particle are updated by the following formula (10) and (11)

$$v_{id}^{t+1} = \omega v_{id}^t + c_1 r_1 \left(P_{id} - x_{id}^t\right) + c_2 r_2 (G_{id} - x_{id}^t) \tag{10}$$

$$x_{id}^{t+1} = x_{id}^t + v_{id}^{t+1} \tag{11}$$

The velocity vector of a particle satisfies the constraint condition (12) - Lipschitz condition of Dynamic System

$$\left\| v_{id}^{t+1} \right\| \leq V_{max} \tag{12}$$

where $\omega$ is inertia weight which can dynamically adjust the particle's velocity, making it tend to converge locally. $v_{id}^t$ and $x_{id}^t$ are the velocity and position of particle $i$ in $d$ dimensional after making $t$ iterations. $c_1$ and $c_2$ are the acceleration factors. They are not negative and control the speed at which each particle moves to position $P_{id}$ and $G_{id}$. $r_1$ and $r_2$ are random numbers and subject to the uniform distribution on interval $[0, 1]$. $v_{id} \in [-v_{max}, v_{max}]$. $v_{max}$ is the maximum velocity, which can be set by the user according to the actual situation.

### 4.1. Improvement of inertia weight

As a coefficient of linear variation, inertia weight $\omega$ is introduced into PSO by Shi and Eberhart [12]. It can balance global search and local search. A larger inertia weight focuses on global searching ability, whereas a smaller one is biased to local searching. The appropriate inertia weight can get the optimal solution with the least iterations. Shi believes that an excellent PSO should have larger inertia weight value at the beginning, ensuring the ability of global search. Later as the time increases, it should be smaller so that the ability of local search can be improved. The omega value of classic inertia weight decreases linearly from 0.9 to 0.4.

Supposing $I_{max}$ is the maximum number of iteration, $\omega_{max}$ is the maximum inertia weight value and $\omega_{min}$ is the minimum. Then the formula of the inertia weight $\omega_i$ after making iterations $i$ times is described by

$$\omega_i = \omega_{max} - \frac{\omega_{max} - \omega_{min}}{I_{max}} i \tag{13}$$

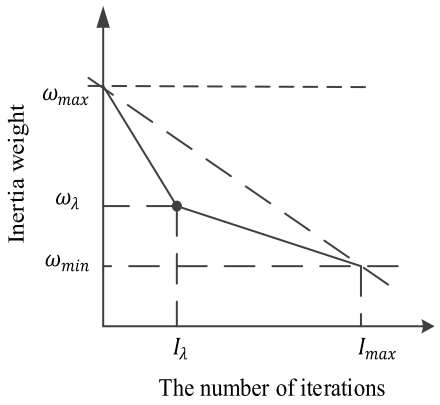



**Fig. 1.** Piecewise weighting function.

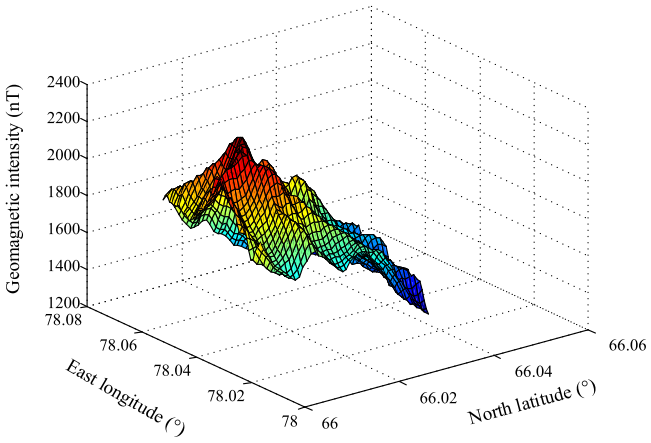



**Fig. 2.** Three dimensional view of geomagnetic intensity.

or

$$\omega_i = \omega_{max} - \frac{\omega_{max} - \omega_{min}}{I_{max} - 1}(i - 1) \tag{14}$$

The constraint of inertia weight during the initialization contributes to quickly locate the global optimum solution. To quickly reduce the inertia weight and make the algorithm into local search phase [13] after a few iterations, the linear variation process is divided into two parts. A parameter named $\omega_\lambda$ is introduced to the process, as shown in Fig. 1.

Its description is given as

$$\begin{cases} \omega_i = \omega_{max} - \dfrac{\omega_{max} - \omega_\lambda}{I_\lambda - 1}(i - 1) & i \le I_\lambda \\ \omega_i = \omega_\lambda - \dfrac{\omega_\lambda - \omega_{min}}{I_{max} - 1}(i - 1) & i > I_\lambda \end{cases} \tag{15}$$

where $I_\lambda$ is the number of iterations in transition. This function can be used to switch global search and local search according to the actual situation [14], effectively improving the flexibility of PSO.

### 4.2. Improvement of encoding mode

The route planning based on traditional PSO takes the geographic coordinates of nodes as particles, and the temporal information is regarded as the particle in the search space [15,16]. The roughing route line is $L_0 = \{S \quad V_1 \quad V_2 \quad \cdots \quad E\}$, while S is the initial node and E is the target node. $V_i$ is the terminal and each terminal is corresponding to $S(x_0, y_0)$, $V_1(x_1, y_1)$, $\cdots$, $E(x_D, y_D)$. If the initial node and the target node are unchanged [17], the roughing route can be adjusted according to the constraint condition. Then the new route is $L'_0 = \{S \quad V'_1 \quad V'_2 \quad \cdots \quad E\}$, which is composed of $V'_i(x'_i, y'_i)$. The adjustment strategy is as follows

$$\begin{cases} x'_i = x_i + t_i(x_{i+1} - x_{i-1}) \\ y'_i = y_i + t_i(y_{i+1} - y_{i-1}) \end{cases} \quad i = 1, 2, \ldots D - 1 \tag{16}$$

The temporal information $t_i$ is used as a particle and its range is $[0, 1]$. This operation can make the route evaluation function minimum. The new route is obtained by searching around the roughing route, rather than in the whole map. So the efficiency is improved.

## 5. Simulation experiment and analysis

The geomagnetic intensity data of a certain sea area is simulated, such as the three-dimensional view shown in Fig. 2. The average geomagnetic intensity is 1831.4016 nT. The minimum is 1344 nT while the maximum is 2329 nT. The geomagnetic intensity goes from 60.0000N to 66.0357N, from 78.0000E to 78.0523E. The data is stored in the shape of grid, whose number is 44*44 and spacing is 0.00083 degrees about 90 m.

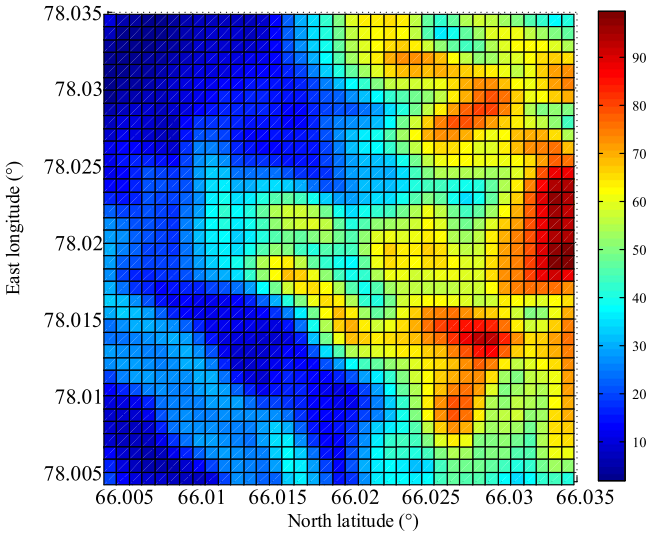



**Fig. 3.** Geomagnetic standard deviation diagram.

The three-dimensional graph is transformed into a two-dimensional graph, and the standard deviation of each point is displayed as the color block diagram as Fig. 3.

The size of local feature window is 6 X 6. The threshold of local geomagnetic standard deviation and correlation coefficient are as follows

$$\sigma > 4\sigma_c \cap r_\lambda < 0.7 \cap r_\phi < 0.7$$

where $\sigma$ is the real standard deviation, $\sigma_c$ is the standard deviation set by magnetometer, $r_\lambda$ is the roughness of the longitude direction, $r_\phi$ is the roughness of the latitude direction.

According to the formula (1), the size of the suitability of each grid can be calculated. Fig. 4 is a map of the adaptation area to be matched. The black area indicates the non-matching area, while the white area indicates the matching area. The horizontal axis represents latitude and the vertical axis represents longitude. Thus it can be calculated that there are 910 matching areas and 690 non-matching areas.

To explore the optimal path from S to E in Fig. 4, firstly, Dijkstra algorithm is used to find the rough route as the primary route of PSO. To save the space, the local window size is 6 X 6. Then the route whose size is 5 X 5 is also calculated, which is compared with the front. It can be concluded that the space is smaller when the local window size is 6 X 6, and the running time is shorter. As shown in Fig. 5, the black line is the roughing track whose size is 5 X 5, and the size of the blue

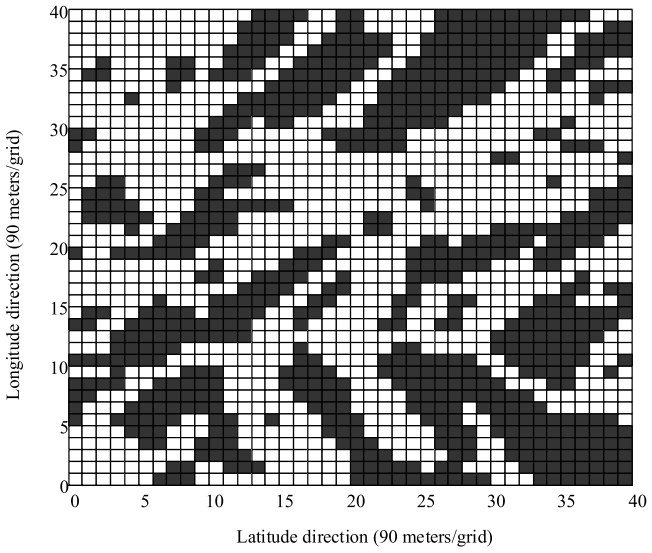



**Fig. 4.** The adaptation area to be matched.

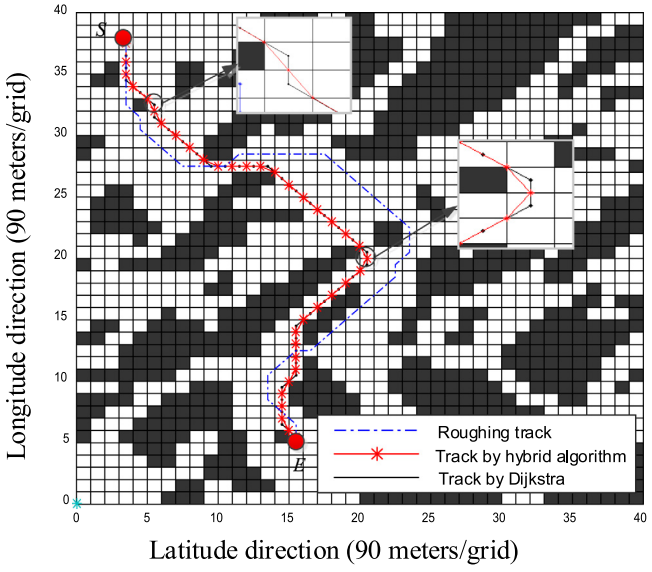



**Fig. 5.** The composite route schematic diagram.. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

one is 6 X 6. Taking the blue route as the initial route and use hybrid algorithm to optimize, then the red line can be obtained.

As can be seen from Fig. 5, the red route obtained by hybrid algorithm is significantly more optimized than the blue rough route obtained only by Dijkstra algorithm. Moreover, the hybrid algorithm optimizes the corners from the view of the landmark, which makes its length shorter than that obtained only by Dijkstra algorithm when the local window size is 5 X 5.

Fig. 6 is a comparison about the fitness between standard PSO and hybrid PSO when the local window size is 6 X 6. The graph (a) represents the hybrid algorithm, while the graph (b) represents the standard. It can be seen that the initial standard PSO fitness function is larger, because its search space is the entire zone. As the number of iterations increases, two algorithms all achieve the global optimum, but the hybrid algorithm requires fewer times. The hybrid algorithm guarantees local optimization and ensures global optimization at the same time. It can effectively avoid the non-matching areas. The effectiveness of hybrid algorithm based on Dijkstra algorithm and PSO is proved.

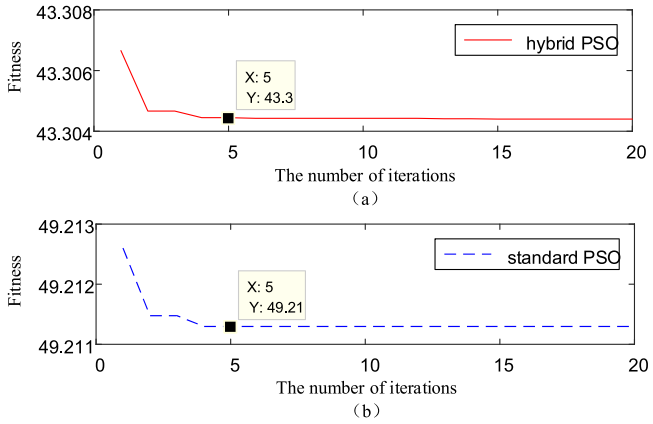



**Fig. 6.** Contrast chart about fitness.

## 6. Conclusion

With the characteristics about geomagnetic navigation environment, the constraints of making vehicle's route planning are necessary to explore, such as geomagnetic map characteristics, the maximum steering angle, obstacle constraints and time constraints. By analyzing the operational environment characteristics of the underwater vehicle path planning, the environmental constraints models are built and normalized, then, the path planning environment constrained cost model is established. Based on relationship among parameters of the environment constrained cost model, a new combined route planning algorithm by using heuristic and random search algorithm can optimize path planning for geomagnetic field navigation. The primary route can be obtained by Dijkstra algorithm in the geomagnetic distribution area, and the primary trajectory can be optimized by PSO algorithm with improved weights and coding. Different inertia weights can make the fitness function reach the global minimum, but the fitness function with piecewise weights is the fastest to converge to the global minimum. Taking the time information as the particle can minimize the search range of evaluation function, then improve the search efficiency. Compared with the standard PSO, the combined algorithm has better global search ability and efficiency, and can acquire the optimal value of local and global fitness. With path planning methods for geomagnetic field navigation by using heuristic and random search algorithm, the non-matching navigation areas can be effectively avoided, and efficient path planning functions can be achieved in underwater geomagnetic matching navigation application.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## CRediT authorship contribution statement

**Yan Xu:** Conceptualization, Methodology. **Guofei Guan:** Resources, Investigation. **Qingwu Song:** Data curation, Visualization. **Chao Jiang:** Software, Validation. **Lihui Wang:** Project administration, Writing - original draft, Writing - review & editing.

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